

COMPLETE DEVOPS FLOW

Linux → Networking → Bash → AWS → Git → Docker → Maven → SonarQube → Trivy
→ Jenkins → Kubernetes → Terraform → Helm → Argo CD → Prometheus → Grafana

1. LINUX

<https://youtu.be/e01GGTKmtpc?si=jlFdOEVTAbwuexN7>

<https://youtu.be/Byx4sgLR88E?si=sskVx3WIXtErMN8F>

Tool / Topic	What is it?	Why / Use
Linux	Open-source operating system	Used to run most DevOps servers and infrastructure
File System	Linux directory structure	Used to organize and manage files
SSH	Secure remote access protocol	Used to connect to remote servers
systemd	Linux service manager	Used to start, stop and manage services
systemctl	Command to manage systemd services	Used to control services
journalctl	Linux log viewer	Used to troubleshoot system and service logs
Cron	Job scheduler	Used to automate tasks at scheduled times
chmod	Permission management command	Used to control file access
chown	Ownership management command	Used to change file ownership
grep	Text-search command	Used to search logs and files
awk	Text-processing tool	Used to extract and process data
sed	Stream editor	Used to modify text automatically
top	Process monitoring command	Used to check CPU and memory usage
df	Disk-space command	Used to check filesystem disk usage
du	Directory-size command	Used to find which directories consume space

2. NETWORKING

<https://youtu.be/xN3BKHji12I?si=v5dY88FQLNfjV1qF>

Tool / Topic	What is it?	Why / Use
IP Address	Unique network address	Used to identify a system on a network
CIDR	IP address range notation	Used to define network and subnet ranges
Subnet	Smaller network inside a network	Used to divide networks logically
Gateway	Network exit point	Used to communicate with other networks
DNS	Domain Name System	Converts domain names into IP addresses
TCP	Connection-oriented protocol	Used for reliable communication
UDP	Connectionless protocol	Used where speed is preferred over reliability
Port	Logical communication endpoint	Used to identify network services
HTTP	Web communication protocol	Used for application/web traffic
HTTPS	Secure HTTP	Used to encrypt web traffic
NAT	Network Address Translation	Used to allow private systems to access external networks
Firewall	Network traffic control	Used to allow or block traffic
Load Balancer	Distributes traffic across servers	Used for scalability and high availability
Reverse Proxy	Server between client and backend	Used for routing, security and load balancing
SSL/TLS	Encryption protocol	Used to secure network communication

3. BASH SCRIPTING

https://youtu.be/9Xl1ZTk3BQw?si=Q920C5oE_r6MfBY6

<youtu.be/TtGM9GfBuok?si=yyE8br7cBeeUe9q5>

Topic	What is it?	Why / Use
Bash	Linux command-line shell	Used to execute commands and automate tasks
Variable	Stores a value	Used to make scripts dynamic
If/Else	Conditional logic	Used to make decisions in scripts
Loop	Repeats commands	Used to automate repetitive tasks
Function	Reusable block of code	Used to organize scripts
\$?	Previous command exit status	Used to check command success/failure
\$1	First script argument	Used to pass input to scripts
\$@	All script arguments	Used to process multiple inputs
Cron + Bash	Scheduled shell automation	Used for recurring server tasks

4. AWS

https://youtu.be/N4sJj-SxX00?si=mEGDFj_bNxzQs4jh

https://youtube.com/playlist?list=PL6XT0grm_TfgtwtwUit305qS-HhDvb4du

Service	What is it?	Why / Use
IAM	Identity and Access Management	Used to control who can access AWS resources
EC2	Virtual server service	Used to run applications and servers
VPC	Virtual private network in AWS	Used to create isolated cloud networking

Subnet	Network segment inside VPC	Used to separate public and private resources
Internet Gateway	VPC internet connection	Used to provide internet access to public resources
NAT Gateway	Outbound internet gateway for private subnets	Used to let private servers access the internet without public IPs
Route Table	Controls network routing	Used to determine where network traffic goes
Security Group	Stateful virtual firewall	Used to control inbound and outbound instance traffic
NACL	Stateless subnet firewall	Used to control subnet-level traffic
S3	Object storage service	Used to store files, backups and static data
EBS	Block storage for EC2	Used to provide persistent disk storage
AMI	EC2 machine image	Used to create servers from predefined configurations
ALB	Application Load Balancer	Used to distribute HTTP/HTTPS traffic
Auto Scaling	Automatically adjusts EC2 capacity	Used to handle changing application traffic
Route 53	AWS DNS service	Used for domain name resolution and routing
ACM	AWS Certificate Manager	Used to manage SSL/TLS certificates
RDS	Managed relational database	Used to run databases without managing the server
ECR	Elastic Container Registry	Used to store Docker images
EKS	Managed Kubernetes service	Used to run Kubernetes clusters on AWS
CloudWatch	AWS monitoring service	Used to monitor resources, logs and metrics
CloudTrail	AWS API activity logging	Used for auditing and tracking AWS actions
WAF	Web Application Firewall	Used to protect applications from web attacks
GuardDuty	Threat detection service	Used to detect suspicious AWS activity
Secrets Manager	Secret storage service	Used to securely store passwords, API keys and credentials

5. GIT & GITHUB

<https://youtu.be/AB3J8ufDYHQ?si=Os1QojjS5pyhux56>

<https://youtu.be/uwyP5SNIQAU?si=jUdNWHWn4wrRl5Dg>

Tool / Topic	What is it?	Why / Use
Git	Distributed version control system	Used to track source-code changes
GitHub	Git repository hosting platform	Used to collaborate and store code remotely
Repository	Project tracked by Git	Used to manage project source code
Branch	Separate line of development	Used to develop features safely
Commit	Snapshot of changes	Used to save changes in Git history
Merge	Combines branches	Used to integrate completed work
Rebase	Reapplies commits on another base	Used to maintain cleaner Git history
Stash	Temporarily stores changes	Used when switching work without committing
Fetch	Downloads remote changes	Used to inspect remote updates without merging
Pull	Fetch + integrate remote changes	Used to update local code
Push	Uploads local commits	Used to send code to GitHub
Tag	Named reference to a commit	Used to mark releases/versions
Revert	Creates a new commit that undoes changes	Used to safely undo production changes
Reset	Moves branch history	Used to modify local commit history
Cherry-pick	Applies one specific commit	Used to move a particular change between branches
Reflog	Records reference movements	Used to recover lost commits
Pull Request	Request to merge code	Used for code review and controlled integration

.gitignore	Git ignore rules	Used to prevent unwanted files from being committed
SSH Authentication	Key-based Git authentication	Used for secure GitHub access without passwords

6. DOCKER

<https://youtu.be/OhnTMWmfTBE?si=soUKLFOKNF6luneU>

<https://youtube.com/playlist?list=PL8p2I9GklV47v6WZTjHAqdsHxpTlpjRwn>

Tool / Topic	What is it?	Why / Use
Docker	Container platform	Used to package and run applications consistently
Image	Read-only application package	Used as the template for containers
Container	Running instance of an image	Used to run applications in isolation
Dockerfile	Instructions for building an image	Used to automate image creation
Docker Engine	Container runtime	Used to build and run containers
Docker Network	Container communication layer	Used to connect containers
Docker Volume	Persistent container storage	Used to keep data after containers restart
Docker Compose	Multi-container tool	Used to run multiple services together
Docker Registry	Image storage system	Used to store and distribute images
Docker Hub	Public Docker registry	Used to share and download container images
ECR	AWS container registry	Used to securely store images for AWS workloads

7. MAVEN

<https://youtu.be/-KqyFvEJYz0?si=SY6VqWc7Kc3a1sMb>

Tool / Topic	What is it?	Why / Use
Maven	Java build and dependency management tool	Used to build and package Java applications
pom.xml	Maven project configuration file	Used to define dependencies, plugins and project settings
Dependency	External library required by application	Used to reuse existing functionality
Maven Lifecycle	Standard build process	Used to automate compile, test and package operations
Maven Repository	Storage for dependencies/artifacts	Used to download and store project dependencies
Snapshot	Development version	Used for frequently changing builds
Release	Stable version	Used for production-ready artifacts
mvn compile	Compiles source code	Used to verify code compilation
mvn test	Runs tests	Used to validate application behavior
mvn package	Creates JAR/WAR package	Used to prepare application for deployment
mvn deploy	Publishes artifact to repository	Used to share built artifacts

8. SONARQUBE

<https://youtu.be/r2UVTDplUj8?si=p0BE2H7h6QlFrZTq>

Tool / Topic	What is it?	Why / Use
SonarQube	Code quality and security analysis platform	Used to detect code problems before deployment
Bug	Coding error	Used to identify functionality problems

Vulnerability	Security weakness	Used to identify security risks
Code Smell	Maintainability problem	Used to improve code quality
Code Coverage	Percentage of code tested	Used to measure test coverage
Technical Debt	Future maintenance effort	Used to track code-quality debt
Quality Gate	Set of quality conditions	Used to decide whether code can proceed
SonarQube Scanner	Code-analysis client	Used to send code for SonarQube analysis

9. TRIVY

https://youtu.be/dwce6YL9N9Q?si=oPPPvjSLDa_pxT2T

Tool / Topic	What is it?	Why / Use
Trivy	Security vulnerability scanner	Used to detect vulnerabilities
Image Scan	Scans container images	Used to find vulnerable packages in images
Filesystem Scan	Scans local files	Used to identify vulnerabilities in project dependencies
Config Scan	Scans configuration files	Used to detect insecure configurations
Repository Scan	Scans source repositories	Used to identify security issues in code/dependencies
Kubernetes Scan	Scans Kubernetes resources	Used to identify Kubernetes security problems
CVE	Common Vulnerabilities and Exposures identifier	Used to track known security vulnerabilities
Severity	Vulnerability risk level	Used to prioritize security fixes

10. JENKINS

<https://youtu.be/XaSdKR2fOU4?si=PDGlWA9FMqazaUiL>

https://youtu.be/NVaP8qtLm6Q?si=SIMoVVz6Fure8_a1

Tool / Topic	What is it?	Why / Use
Jenkins	CI/CD automation server	Used to automate build, test and deployment
Pipeline	Automated sequence of stages	Used to define CI/CD workflow
Jenkinsfile	Pipeline-as-code file	Used to store CI/CD pipeline configuration in Git
Stage	Logical pipeline section	Used to organize build steps
Step	Individual pipeline action	Used to execute commands/tasks
Agent	Machine that executes Jenkins jobs	Used to run pipeline workloads
Credential	Stored secret in Jenkins	Used to securely access Git, AWS, Docker etc.
Webhook	Event notification from GitHub	Used to automatically trigger Jenkins
Plugin	Jenkins extension	Used to add integrations and functionality
CI	Continuous Integration	Used to automatically build and test code
CD	Continuous Delivery/Deployment	Used to automate application delivery

JENKINS PIPELINE

GitHub – Jenkins – Maven – SonarQube – Trivy – Docker – ECR -- Kubernetes

11. KUBERNETES

<https://youtube.com/playlist?list=PLY63ZQr2Y5BHkJJhwPjJuJ41Clv3m7Ru>

https://youtu.be/rBeyHDKLVqM?si=a0XiQ-bDXgnxB8_c

Tool / Topic	What is it?	Why / Use
Kubernetes	Container orchestration platform	Used to deploy, scale and manage containers
Cluster	Group of Kubernetes resources	Used to run containerized applications

Control Plane	Kubernetes management layer	Used to control and manage the cluster
Worker Node	Machine running workloads	Used to execute application containers
Pod	Smallest Kubernetes deployable unit	Used to run one or more containers
Deployment	Manages stateless Pods	Used for application deployment and updates
ReplicaSet	Maintains desired Pod count	Used to ensure required replicas are running
Service	Stable network endpoint	Used to expose Pods to other services/users
Namespace	Logical cluster partition	Used to separate workloads and resources
ConfigMap	Non-sensitive configuration storage	Used to provide application configuration
Secret	Sensitive configuration storage	Used to store passwords, tokens and keys
Ingress	HTTP/HTTPS routing resource	Used to route external traffic to services
HPA	Horizontal Pod Autoscaler	Used to automatically scale Pods
StatefulSet	Controller for stateful workloads	Used for databases and applications requiring stable identity
DaemonSet	Runs a Pod on selected nodes	Used for node-level agents such as monitoring/logging
Job	Runs a task to completion	Used for one-time workloads
CronJob	Scheduled Kubernetes Job	Used for recurring tasks
PV	Persistent storage resource	Used to provide storage to workloads
PVC	Request for persistent storage	Used by Pods to request storage
RBAC	Role-based access control	Used to control Kubernetes permissions
Liveness Probe	Checks application health	Used to restart unhealthy containers
Readiness Probe	Checks whether app can receive traffic	Used to prevent traffic going to unready Pods
Resource Request	Minimum resource requirement	Used for Kubernetes scheduling
Resource Limit	Maximum resource allowed	Used to prevent excessive resource consumption

12. TERRAFORM

https://youtu.be/S9mohJI_R34?si=MOL91o1VscITcl8B

Tool / Topic	What is it?	Why / Use
Terraform	Infrastructure as Code tool	Used to create and manage infrastructure using code
Provider	Plugin connecting Terraform to a platform	Used to manage AWS, Azure, Kubernetes etc.
Resource	Infrastructure object managed by Terraform	Used to create servers, networks and services
Variable	Input value	Used to make Terraform configurations reusable
Output	Displays resource information	Used to expose useful infrastructure values
Data Source	Reads existing infrastructure information	Used to reference resources not managed by Terraform
Module	Reusable Terraform configuration	Used to standardize infrastructure
State	Terraform's record of infrastructure	Used to track managed resources
Backend	Location where state is stored	Used for centralized/remote state management
Remote State	State stored remotely	Used for team collaboration
State Locking	Prevents simultaneous state changes	Used to avoid Terraform state corruption
Workspace	Separate Terraform state/environment	Used to manage multiple environments
terraform init	Initializes Terraform	Used to download providers and configure backend
terraform plan	Shows planned changes	Used to preview infrastructure changes
terraform apply	Applies configuration	Used to create/update infrastructure
terraform destroy	Removes managed infrastructure	Used to delete resources

13. HELM

<https://www.youtube.com/live/6Yv3eGHFv2A?si=81IQ4Vxob52Z2zOe>

Tool / Topic	What is it?	Why / Use
Helm	Kubernetes package manager	Used to package and deploy Kubernetes applications
Helm Chart	Collection of Kubernetes templates	Used to package an application
Chart.yaml	Chart metadata file	Used to define chart information
values.yaml	Configuration values file	Used to customize deployments
Templates	Dynamic Kubernetes manifests	Used to generate Kubernetes YAML
Helm Repository	Storage for Helm charts	Used to distribute charts
Release	Installed instance of a Helm chart	Used to track a deployed application
helm install	Installs a chart	Used to deploy applications
helm upgrade	Updates a release	Used to deploy new application versions
helm rollback	Reverts a release	Used to recover from failed deployments

14. ARGO CD

https://youtu.be/iGJxNtjj_VE?si=a5G0iTqDN4f3UL1J

Tool / Topic	What is it?	Why / Use
Argo CD	GitOps continuous delivery tool	Used to deploy Kubernetes applications from Git
GitOps	Git-based operational model	Used to manage infrastructure/application desired state
Application	Argo CD deployment definition	Used to connect Git with Kubernetes

Repository	Git source configured in Argo CD	Used as the source of deployment manifests
Sync	Applies Git state to Kubernetes	Used to make cluster match Git
Auto Sync	Automatic synchronization	Used to deploy Git changes automatically
Self Heal	Automatically corrects drift	Used to restore Kubernetes to Git-defined state
Health Status	Application health state	Used to monitor deployment health
Sync Status	Git vs cluster state	Used to detect configuration differences
Rollback	Return to previous application version	Used to recover from bad deployments
Multi-Cluster	Manage multiple Kubernetes clusters	Used to deploy applications across environments/clusters

GITOPS FLOW

Developer -- Git Push -- Git Repository -- Argo CD -- Sync -- Kubernetes / EKS -- Production

15. PROMETHEUS

<https://youtu.be/DXZUunEeHqM?si=aoRfLLOjA0FtRd>

Tool / Topic	What is it?	Why / Use
Prometheus	Metrics monitoring system	Used to collect and query infrastructure/application metrics
Metric	Numerical measurement	Used to understand system performance
Target	Endpoint monitored by Prometheus	Used to identify metric sources
Exporter	Converts metrics into Prometheus format	Used to expose metrics from systems
Node Exporter	Linux system metrics exporter	Used to monitor CPU, RAM, disk and network

kube-state-metrics	Kubernetes object metrics exporter	Used to monitor Pods, Deployments and other objects
PromQL	Prometheus query language	Used to query and analyze metrics
Label	Key-value metadata on metrics	Used to filter and group metrics
Recording Rule	Precomputed PromQL result	Used to improve performance of repeated queries
Alerting Rule	Condition that triggers an alert	Used to detect problems automatically
AlertManager	Prometheus alert management component	Used to route and notify alerts

MONITORING FLOW

Server / Kubernetes – Exporter – Prometheus – PromQL -- AlertManager

16. GRAFANA

<https://youtu.be/rt55FcSyl4o?si=Jtw-xrMk5rgcaOYO>

Tool / Topic	What is it?	Why / Use
Grafana	Observability visualization platform	Used to visualize metrics, logs and alerts
Data Source	Backend providing data	Used to connect Grafana to Prometheus, Loki etc.
Dashboard	Collection of visualizations	Used to monitor systems from one place
Panel	Individual visualization	Used to display a specific metric/query
Query	Request for monitoring data	Used to retrieve required metrics/logs
Variable	Dynamic dashboard value	Used to filter dashboards by cluster, node, namespace etc.
Alert	Condition-based notification	Used to notify when systems have problems
Contact Point	Alert destination	Used to send notifications to email, Slack etc.
Notification Policy	Alert routing rules	Used to control where alerts are sent

GRAFANA FLOW

Prometheus – Grafana – Dashboard – Visualization – Alert -- Notification

DEVOPS TOOLS — WHAT & WHY

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Linux → Operating System → Used to manage DevOps servers and applications.

Networking → System Communication → Used to connect servers, applications, users and cloud resources.

Bash → Shell Scripting → Used to automate Linux tasks and repetitive operations.

AWS → Cloud Platform → Used to build, host and manage scalable cloud infrastructure.

Git → Version Control → Used to track code changes and manage source code history.

GitHub → Git Repository Platform → Used to store code, collaborate and manage Pull Requests.

Docker → Container Platform → Used to package applications with dependencies and run them consistently.

Maven → Java Build Tool → Used to compile, test and package Java applications.

SonarQube → Code Quality Tool → Used to detect bugs, code smells, vulnerabilities and improve code quality.

Trivy → Security Scanner → Used to scan code, Docker images and configurations for vulnerabilities.

Jenkins → CI/CD Automation Tool → Used to automate build, test, security scanning and deployment pipelines.

Kubernetes → Container Orchestration Platform → Used to deploy, scale, manage and self-heal containers.

Terraform → Infrastructure as Code Tool → Used to create and manage AWS/cloud infrastructure using code.

Helm → Kubernetes Package Manager → Used to package, configure and deploy Kubernetes applications easily.

Argo CD → GitOps CD Tool → Used to automatically deploy Kubernetes applications from Git.

Prometheus → Monitoring & Metrics Tool → Used to collect and query infrastructure and application metrics.

Grafana → Visualization & Dashboard Tool → Used to visualize metrics, logs and alerts in dashboards.